Fiscal Year:	FY 2010	**Task Last Updated:**	FY 09/24/2009
PI Name:	Almeida, Eduardo A C Ph.D.		
Project Title:	The Effects of Microgravity on the Tissue Regenerative Potential of Stem Cells		
Division Name:	Space Biology		
Program/Discipline:	SPACE BIOLOGY		
Program/Discipline--Element/Subdiscipline:	SPACE BIOLOGY--Cellular and molecular biology		
Joint Agency Name:		**TechPort:**	No
Human Research Program Elements:	None		
Human Research Program Risks:	None		
Space Biology Element:	None		
Space Biology Cross-Element Discipline:	None		
Space Biology Special Category:	None		
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Comments:			
Project Type:	Flight	**Solicitation / Funding Source:**	2008 Space Biology--Microbial, Plant, Cell Biology: NNH08ZTT003N
Start Date:	10/01/2009	**End Date:**	05/31/2012
No. of Post Docs:		**No. of PhD Degrees:**	
No. of PhD Candidates:		**No. of Master' Degrees:**	
No. of Master's Candidates:		**No. of Bachelor's Degrees:**	
No. of Bachelor's Candidates:		**Monitoring Center:**	NASA ARC
Contact Monitor:	Sun, Sidney C	**Contact Phone:**	650-604-4835
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Flight Program:	Shuttle/ISS		
Flight Assignment:			
Key Personnel Changes/Previous PI:			
COI Name (Institution):			
Grant/Contract No.:			
Performance Goal No.:			
Performance Goal Text:			
	NOTE that this is the Flight portion of project entitled, "The Role of Artificial Gravity in Promoting Tissue-Regenerative Matrix-Integrin-Kinase Cell Signaling." See that project for annual reports as projects were combined in April 2011 for reporting purposes (Ed. 7/11/2011) Our broad working hypothesis is that in space, mechanical unloading caused by microgravity can alter the growth and regenerative health of a broad range of mammalian tissues. Conversely, we hypothesize that gravity on earth activates the extracellular Matrix-Integrin-Kinase (MIK) pathway, a key cell-signaling pathway responsible for maintaining normal regenerative tissue growth. Specifically we hypothesize that somatic stem cells responsible for tissue regeneration may require the mechanical stimulation of gravity to proliferate and regenerate tissues at normal rates, and that microgravity may slow down tissue		

Task Description:	regeneration. We also hypothesize that the MIK pathway plays a key role in epigenetic reprogramming of stem cells into a state of differentiation and tissue regeneration. In support of these hypotheses, phenomenological findings from our recent spaceflight experiments with amphibian (newt) tail regeneration during the Foton M2 and Foton M3 missions, suggest that spaceflight significantly retards tissue regeneration. Ongoing ground studies in our laboratory supported by the companion ground component of the NNH08ZTT003N NRA aim to test the hypotheses that 1), gravity-dependent mechanical loading stimulates cell proliferation and differentiation in the newt tail regeneration spaceflight model via the MIK signaling pathway, and 2), that gravity-dependent mechanical loading activates the MIK signaling pathway in stem cells, and increases their differentiation and proliferation. To test the microgravity component of our hypotheses we propose a new spaceflight experiment with a cell-based regeneration model using embryoid body (EB) differentiation from embryonic stem (ES) cells (ES to EB model). Our aims are: AIM 1. To test the hypothesis that microgravity-dependent mechanical unloading decreases tissue differentiation and growth in the ES to EB model of regeneration. AIM 2. To test the hypothesis that microgravity alters normal epigenetic patterns of gene silencing during tissue differentiation and growth in the ES to EB model of tissue regeneration. A major potential benefit of successfully concluding this study will be to determine whether microgravity plays a role in reducing somatic stem cell-based tissue regeneration. In addition we will determine whether epigenetic molecular mechanisms involved in reprogramming stem cells for regenerative tissue-differentiation and growth are also affected by microgravity. Both of these outcomes would help define risk for long-term tissue degeneration in space, and possibly provide a rationale to develop novel countermeasures.
Rationale for HRP Directed Research:	
Research Impact/Earth Benefits:	
Task Progress:	New project for FY2009. NOTE that this is the Flight portion of project entitled, "The Role of Artificial Gravity in Promoting Tissue-Regenerative Matrix-Integrin-Kinase Cell Signaling." See that project for annual reports as projects were combined in April 2011 for reporting purposes (Ed. 7/11/2011).
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